

Night-bombing operations involved many hazards, even apart from enemy flak and fighters. A heavily laden Lancaster or Halifax was not easy to get into the air, and difficult night landings were legion – even if the bomber was undamaged, it was still not simple to land at night when airfields could not be lit because of possible enemy intruders. The weather was frequently a source of anxiety. Bombers would ice up dangerously, and many pilots' worst nightmare stories were of being caught in thunderclouds, which could send a Lancaster whirling out of control like a leaf in a gale. Flying in a bomber

INTERROGATION SCENE

RAF Flying Officer G. Dunbar (second from left) leads a group of officers through a debriefing following a night raid on Berlin in late November 1943.

stream was in itself a considerable feat of airmanship, with the risk of a midair collision always present. Over the target, there was often the risk of being hit by bombs falling from another bomber at a higher altitude. Yet pilots also recalled moments of relief and even joy flying these mighty aircraft through the night – skimming along the top of a thick cloud layer under the stars, or relaxing with a cigarette on the return flight across the North Sea, the autopilot on and the oxygen switched off, looking forward to a breakfast of bacon and eggs.

Navigational aids

The development of Bomber Command's night offensive against German cities and industrial areas through 1942 and 1943 was aided by major advances in equipment and operational tactics. First, they had much better aircraft in the four-engined Halifax and Lancaster bombers, capable of carrying an impressive bombload into the heart of Germany. Second, there were

navigational aids that allowed them to match the night-bombing accuracy that the Luftwaffe had achieved during the Blitz.

The first of these was Gee: a grid of radio beams projected over Germany from sites in Britain were picked up by aircraft receivers, allowing navigators to plot their progress accurately. This was enough to ensure that the average bomber crew could find a German city in the dark. Next came Oboe: one beam guided a bomber to its target, while a series of crossbeams told the crew when they were drawing near to their goal and when to release their bombs. Finally, in 1943, there was an airborne lookdown radar, H2S, which improved chances of identifying targets on the ground even at night and in heavy cloud (see page 242).

Oboe and H2S were mostly used by elite squadrons of the Pathfinder Force (PFF). Flying Wellingtons or Stirlings (and, later, de Havilland Mosquitos), their job was to mark targets accurately with incendiaries for the less skillful pilots

THE BATTLE OF HAMBURG

IN THE CAT-AND-MOUSE GAME that was night war over Germany, a new move giving a temporary advantage could have devastating results. This is what happened over the port city of Hamburg in late July 1943. The RAF had a new trick called "window," the scattering of strips of aluminum foil out of the bomber's flare chute to confuse German radar. Without radar, the antiaircraft guns, the night fighters, and their controllers were blinded and let the bombers through. During the short time it took the German defenders to adapt to the new situation, a series of raids by the RAF laid Hamburg to waste.

On the night of July 27–28, 735 bombers dropped 2,564 tons (2,326 tonnes) of explosives and incendiaries in little over an hour, starting fires that killed over 40,000 people. The devastation of Hamburg represented the ideal to which Bomber Command aspired – an act of destruction on such a scale that, if repeated night after night, might have made the Germans think seriously about ending the war. But the bombers could not achieve this. They were not to repeat such destruction until Dresden in 1945.

STRATEGIC AIR OFFENSIVE

An RAF Lancaster is silhouetted against flames, smoke, and flak during a Bomber Command attack on Hamburg in 1943.

